

Title page:

Tea consumption and the risks of all-cause, cardiovascular disease, and cancer mortality: a meta-analysis of ~~thirty-eight~~38 prospective cohort data sets

Running title: Tea and mortality

Youngyo Kim¹ and Youjin Je^{2*}

¹ Department of Food and Nutrition/Institute of Agriculture and Life Science, Gyeongsang National University, Jinju, South Korea

² Department of Food and Nutrition, Kyung Hee University, Seoul, South Korea

Corresponding author*

Youjin Je, ScD

Department of Food and Nutrition,

Kyung Hee University,

26 Kyunghee-daero, Dongdaemun-gu,

Seoul 02447, South Korea

Tel: +82-2-961-0258

Fax: +82-2-961-0538

Email: youjinje@khu.ac.kr

Funding

This work was supported by the National Research Foundation of Korea, funded by the Ministry of Science, ICT, and Future Planning (~~G~~grant number: NRF-2021R1F1A1050847).

~~F~~The funders had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Abstract: 246 words

Text: 3396 words

References: 60

Tables/Figures: 2 tables and 7 figures

Abstract

Objectives: Tea consumption has been considered beneficial to human health because tea contains phytochemicals such as polyphenols and theaflavins. We conducted a systematic review and meta-analysis on the association between tea consumption and mortality from all causes, cardiovascular disease (CVD), and cancer to provide a quantitative assessment of current evidence.

Methods: The PubMed, Web of Science, and Scopus databases were searched through April 2024 to identify eligible studies. Random effects models were used to combine study-specific effect estimates (ESs).

Results: A total of 38 prospective cohort data sets (from 27 papers) with 1,956,549 participants were included in this meta-analysis. The pooled ESs of the highest *versus* lowest categories of tea consumption were 0.90 (95% CI, 0.86-0.95) for all-cause mortality, 0.86 (95% CI, 0.79-0.94) for CVD mortality, and 0.90 (95% CI, 0.78-1.03) for cancer mortality. In the dose-response analysis, a nonlinear association was observed. The greatest risk reductions were observed for the consumption of 2 cups/day for all-cause mortality (ES, 0.91; 95% CI, 0.88-0.94) and 1.5 cups/day for cancer mortality (ES, 0.92; 95% CI, 0.89-0.96), whereas additional consumption did not show a further reduction in the risk of death. A plateau was observed for CVD mortality at moderate consumption levels (1.5–3 cups/day), but a sustained reduction in mortality risk was observed at higher intake levels.

Conclusions: Moderate tea consumption (e.g., 1.5–2 cups/day) was associated with lower all-cause, CVD, and cancer mortality compared to no tea consumption. Further well-designed prospective studies are needed for a definitive conclusion.

Keywords: tea, all-cause mortality, CVD mortality, cancer mortality, meta-analysis

Introduction

Tea has been one of the most commonly consumed beverages worldwide for centuries [1]. Tea is a rich source of flavonoids such as catechins, theaflavins, anthocyanins, and flavonols [2]. Flavonoids are known for their potential to reduce the risk of many non-communicable diseases, including cardiovascular disease (CVD) and cancer, due to their anti-inflammatory and antioxidant properties that reduce oxidative stress [3, 4]. In addition, a meta-analysis involving 386,610 participants showed that high flavonoid intake was associated with a lower risk of mortality from CVD and all causes [5].

Because tea is widely consumed globally, its effect on public health can be major even if its effect is minor on an individual level. Many observational studies have explored the associations between tea consumption and the risks of all-cause and cause-specific mortality [6-32]. Some studies found an inverse [7, 17, 21-24, 27, 28, 30] or positive [9, 10] association between tea consumption and the risk of mortality, whereas others did not find a significant association [6, 8, 11-16, 18-20, 25, 26, 29, 31, 32]. To date, several meta-analyses have been conducted to examine the association between tea consumption and risk of mortality, but they included studies in which all participants had a specific disease [33, 34] or did not provide results for cancer mortality [34, 35]. Since then, many large cohort studies have been published, analyzing data from the UK Biobank (n= 498,158) and Asia Cohort Consortium (n=528,504), [27-32]. Recent large cohort studies have indicated that significant associations exist between tea consumption and mortality [27, 28, 30]. Given the results published to date, we speculated about the possible association between tea consumption and mortality if a meta-analysis were conducted on the general population, excluding as many studies as possible that targeted only patients.

Therefore, we performed a systematic review and meta-analysis of prospective cohort studies to provide the latest evidence on the associations between tea consumption and the risk

of mortality from all causes, CVD, and cancer in the general population. Furthermore, a dose-response analysis was conducted to determine trends in mortality risk with increased tea consumption and elucidate appropriate levels of tea consumption to reduce the risk of death.

Material and methods

Data sources and searches

A systematic literature search was performed using the PubMed, ISI Web of Science, and Elsevier Scopus databases to identify eligible studies published in English through April 2024. We used the following keywords in the search: “(tea OR beverage OR beverages) AND (mortality OR death OR fatal).” A manual search that reviewed the included articles' reference lists was also conducted to identify additional eligible studies. This meta-analysis was registered on PROSPERO (CRD42023400484).

Study selection

Studies were included in the current meta-analysis according to the following criteria: (1) the study had a prospective observational design; (2) the exposure of interest was tea consumption; (3) the outcome of interest was mortality from all causes, CVD, and cancer; (4) the researchers reported effect estimates (relative risks [RRs] or hazard ratios [HRs]) with the corresponding 95% confidence intervals (CIs) for each tea-intake group. If 2 different papers had been published on the same cohort data sets, the study with a more extended follow-up period or more participants was included in the meta-analysis. If 2 papers following the same cohort reported deaths from different causes, both were included in the meta-analysis [19, 32]. Studies in which the participants consisted only of patients with a specific disease were excluded.

Data extraction and quality assessment

The relevant data were independently extracted by 2 authors (Y.K. and Y.J.) according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [36]. The following information was collected from each paper: surname of first author, year of publication, cohort name, number of participants and cases, age, sex, geographical region or country, period of follow-up, categories of tea consumption, the effect estimates (ESs [RRs or HRs]) and 95% CIs for each category of tea consumption, and adjusted variables.

Quality assessment of the included studies was performed using the Newcastle-Ottawa quality assessment scale [37]. We assigned a maximum score of 9 points to each study based on the following criteria: selection of participants (0-4 points), comparability of cohorts (0-2 points), and ascertainment of outcomes of interest (0-3 points). Studies with total scores of 0-3, 4-6, or 7-9 points were regarded as low, moderate, or high quality, respectively. Any discrepancies between authors in data extraction and quality assessment were resolved by reviewing and discussing the original papers.

Statistical analysis

The natural logarithm values of each study's ESs (RRs or HRs) were combined using a DerSimonian and Laird [38] random effects model, which considers both within-study and between-study variability. If a study provided separate ESs by the presence or absence of disease history, we used the ESs for the latter [32]. The summary ESs were displayed as forest plots. The Cochran Q test [39] was used to assess heterogeneity among studies, and the I^2 statistic [40] was used to quantify inconsistency. Subgroup analyses stratified by sex, geographic region, follow-up years, and sample size were conducted to investigate the variations in ESs among the studies. To assess how much a specific study may influence inferences, we performed a sensitivity analysis by omitting a single study at a time and deriving pooled ESs. Publication bias was evaluated with the Begg test [41] and the Egger regression

test [42], as well as funnel plot visualizations.

To estimate the study-specific slope lines, a linear dose-response meta-analysis was conducted using the methods proposed by Greenland and Longnecker [43-45]. The median value of tea consumption was assigned to each category based on information presented in the original paper. When the highest category was open-ended, we regarded it as of the same magnitude as the adjacent category. Referring to previous papers, we considered 150 mL of tea consumption to be 1 cup of tea [9, 33]. To investigate a potential nonlinear association between tea consumption and risk of mortality, we used restricted cubic splines with 4 knots at fixed percentiles (5th, 35th, 65th, and 95th percentiles) of aggregated tea consumption. The p-value for non-linearity was computed by testing the null hypothesis that the regression coefficient of the second spline is equal to zero [46]. A 2-tailed p-value of less than 0.05 was considered to indicate statistical significance. All statistical analyses were carried out with Stata 17.0 (StataCorp, College Station, TX, USA).

Results

Study characteristics

A total of 38 prospective cohort data sets (27 papers) including 1,956,549 participants and 218,948 deaths from all causes, 53,234 deaths from CVD, and 56,049 deaths from cancer were eligible for this meta-analysis [6-32] (**Figure 1**). **Table 1** summarizes the characteristics of the included studies. Among them, 7 papers reported all estimates of mortality from all causes, CVD, and cancer [17, 21-23, 25, 26, 30], respectively. Six papers provided estimates of mortality from all causes and CVD [10, 15, 18, 24, 27, 28], and 1 paper reported estimates of mortality from all causes and cancer [9]. The other papers provided mortality from either all causes [6, 8, 13, 16, 31, 32], CVD [7, 11, 12, 19, 20] or cancer only [14, 29]. The follow-up durations varied from 5 years to 23 years, and the mean follow-up time was 12.3 years. The

studies were conducted in Asia [11, 13, 14, 17, 19, 22, 27, 30-32], Europe [7, 9, 10, 12, 18, 26, 28, 29], USA [6, 8, 15, 16, 20, 21, 23, 25], and Oceania [24]. All studies were adjusted for age and smoking, and most were adjusted for alcohol consumption [7, 9-11, 13, 15-19, 21-23, 25-32] and body mass index (BMI) [7-12, 15-17, 19-32]. The quality assessment results ranged from 7 to 9, indicating high quality (Supplementary Table 1).

All-cause mortality

Thirty-three prospective cohort data sets (20 papers) involving 1,825,815 participants and 218,948 deaths examined the association between tea consumption and all-cause mortality [6, 7, 9, 10, 13, 15-18, 21-28, 30-32]. The summary ES for all-cause mortality, comparing the highest and lowest levels of tea consumption, was 0.90 (95% CI, 0.86-0.95), with significant heterogeneity among studies ($I^2=81.6\%$, $p<0.001$) (**Figure 2**). The sensitivity analysis results, excluding 1 study at a time, showed that the pooled ESs ranged from 0.89 (95% CI, 0.85-0.94) to 0.92 (95% CI, 0.88-0.96) (Supplementary Figure 1). The inverse association between tea consumption and the risk of all-cause mortality was slightly stronger in men (ES, 0.87; 95% CI, 0.77-0.98) than in women (ES, 0.90; 95% CI, 0.79-1.01), but the difference was not statistically significant (p for difference = 0.97) (**Table 2**). By geographic region, the strongest inverse association between tea consumption and all-cause mortality was shown in Asia (ES, 0.84; 95% CI, 0.77-0.91), while an insignificant positive association was observed in Europe (ES, 1.12; 95% CI, 0.88-1.42) (p for difference = 0.05). We did not observe significant variation according to follow-up times (p for difference = 0.17) or number of participants (p for difference = 0.87). We found a nonlinear association between tea consumption and all-cause mortality (p for nonlinearity <0.0001) (**Figure 3**). A significant reduction in all-cause mortality was observed with up to 2 cups/day of tea consumption, and additional consumption did not show a further decrease in mortality.

CVD mortality

Thirty-one prospective cohort data sets (18 papers) involving 1,820,381 participants and 53,234 deaths examined the association between tea consumption and CVD mortality [7, 10-12, 15, 17-28, 30]. The summary ES for CVD mortality, comparing the highest and lowest levels of tea consumption, was 0.86 (95% CI, 0.79-0.94), showing significant heterogeneity among studies ($I^2=74.6\%$, $p<0.001$) (**Figure 4**). When the pooled ES was estimated after omitting 1 study at a time, the ESs ranged from 0.85 (95% CI, 0.78-0.92) to 0.88 (95% CI, 0.82-0.95) (Supplementary Figure 2). The inverse association between tea consumption and CVD mortality was similar in men (ES, 0.84; 95% CI, 0.71-0.99) and women (ES, 0.86; 95% CI, 0.75-0.99). By geographic region, the largest reduction in CVD mortality was found in Asia (ES, 0.75; 95% CI, 0.65-0.88), but the difference was not statistically significant (p for all comparisons > 0.1) (**Table 2**). Some evidence for a nonlinear association was found between tea consumption and CVD mortality (p for nonlinearity = 0.0001) (**Figure 5**). Despite the significant nonlinearity showing an identical degree of risk reduction for 1.5–3 cups per day, the ES decreased by 27% with up to 8 cups of tea consumption per day.

Cancer mortality

Twenty-two prospective cohort data sets (11 papers) involving 1,439,658 participants and 56,049 deaths examined the association between tea consumption and cancer mortality [9, 13, 14, 17, 21-23, 26, 29, 30]. The summary ES for cancer mortality, comparing the highest and lowest levels of tea consumption, was 0.90 (95% CI, 0.78-1.03), with significant heterogeneity among studies ($I^2=78.8\%$, $p<0.001$) (**Figure 6**). When we calculated the pooled ES after excluding 1 study at a time, the ESs ranged from 0.87 (95% CI, 0.75-1.00) to 0.94 (95% CI, 0.82-1.07) (Supplementary Figure 3). We found no significant difference by sex (p for

difference = 0.93) or geographic region (p for difference > 0.2 in all comparisons) (**Table 2**). A significant nonlinear association was found between tea consumption and cancer mortality (p for nonlinearity < 0.0001) (**Figure 7**). The most robust inverse association was shown for tea consumption of 1.5–2 cups/day, and a significant reduction in cancer mortality was observed with tea consumption of up to 5.5 cups/day.

Publication bias

No evidence of publication bias was found for tea consumption and all-cause mortality (Begg's $p = 0.97$; Egger's $p = 0.75$), CVD mortality (Begg's $p > 0.99$; Egger's $p = 0.65$), or cancer mortality (Begg's $p = 0.88$; Egger's $p = 0.43$).

Discussion

The current systematic review and meta-analysis of 38 prospective cohort data sets (from 27 papers) with 1,956,549 total participants found inverse associations between tea consumption and all-cause and CVD mortality. People with the highest level of tea consumption had a 10% lower risk of all-cause mortality and a 14% lower risk of CVD mortality than those with the lowest consumption level. The results from the dose-response analyses showed a nonlinear association between tea consumption and mortality from all causes, CVD, and cancer. The most significant reduction in all-cause mortality was observed at 2 cups/day of tea consumption, and further intake did not appear to further lower the risk of death. A plateau in CVD mortality was observed at moderate intake levels (1.5–3 cups/day of tea consumption), but a continued decline in mortality risk was observed at higher intake levels. For cancer mortality, the greatest reduction in mortality risk was observed at the level of consuming 1.5 cups of tea per day, and additional intake did not lower mortality.

Many previous observational studies have reported the association between tea

drinking and the risk of chronic diseases. A recent meta-analysis of 19 prospective cohort studies, with a total of 1,076,300 participants, found that people consuming 4 or more cups of tea per day had a 17% lower risk of type 2 diabetes than people who did not drink tea [47]. For CVD risk, a meta-analysis of 7 prospective cohort studies involving 9,211 cases of coronary heart disease among 772,922 participants indicated that green tea consumers had less risk of coronary heart disease than those who did not consume green tea. The risk was 11% lower for 1 cup/day, 16% lower for 2 cups/day, 15% lower for 3 cups/day, and 12% lower for 4 cups/day [48]. Furthermore, a meta-analysis of 5 prospective cohort studies, including 11,421 stroke cases among 645,393 participants, reported that 900 mL/day of green tea consumption was associated with a 23% reduced risk of stroke [49]. Regarding risk of cancer, meta-analyses of prospective cohort studies showed that high tea consumption was inversely associated with several types of cancer, including glioma and ovarian, liver, bladder, and lymphoid cancer [50-52]. The lower risk of mortality with tea consumption observed in the current study may be due to the inverse association between tea consumption and the risks of many chronic diseases reported in previous studies.

A previous meta-analysis reported 26% lower and 24% lower risks of cardiac death and all-cause death, respectively, per 3-cup increment of tea consumption [34]. Another meta-analysis published in 2015 found that green tea consumption was inversely associated with mortality from CVD and all causes, and black tea consumption was inversely associated with mortality from cancer and all causes [33]. However, in that meta-analysis, when classifying the types of tea, the types of tea were estimated and analyzed according to the region where the study was conducted [33]. A recent meta-analysis from 2020 observed that each 1-cup increase in daily tea intake was associated with 4% and 2% lower risks of mortality from CVD and all causes, respectively [35]. Previous results and the results of our study are consistent in that tea consumption was inversely associated with the risk of CVD and all-cause mortality, and our

findings indicated that consuming 1.5–2 cups of tea daily was associated with lower mortality rates from CVD, cancer, and all causes.

In the subgroup analysis by geographical region, the inverse association between tea consumption and risk of mortality from all causes was stronger in Asia than in Europe. These differences observed across regions may be due to regional differences in the types of tea that are widely consumed. Among the studies included in this meta-analysis, those that reported the type of tea were mostly from Asia, and it was primarily green tea. Although studies conducted in Europe did not report the type of tea, it is generally known that people in Europe and North America mainly consume black tea [1, 53]. Green tea and black tea go through different manufacturing processes, and their composition varies accordingly. The most abundant polyphenols in green tea are catechins such as epicatechin, epicatechin-3-gallate, epigallocatechin, and epigallocatechin-3-gallate (EGCG). Meanwhile, the major polyphenolic components of black tea are theaflavin and thearubigin [53]. These differences in the composition of green tea and black tea may have influenced the difference in their associations with mortality risk. Most of the studies included in this meta-analysis did not report the ES by type of tea. The single study that reported ES by tea types found a significant inverse association between green tea consumption and mortality from all causes and CVD [27]. Specifically, drinking green tea 3 or more times a week was associated with a 21% lower risk of mortality from all causes and a 22% lower risk of mortality from CVD than drinking it fewer than 3 times a week. The researchers did not find a significant association between black tea consumption and mortality risk. It appears that more large-scale cohort studies need to be conducted in the future to determine whether mortality risk varies depending on the type of tea. Another explanation for the differences by geographical region in the association between tea consumption and mortality may be differences in how tea is consumed. In Europe, especially in the UK, it is common to add milk to tea [9]. Proteins in milk can combine with polyphenols

in tea [54], reducing the polyphenols' antioxidant capacity by inhibiting absorption and decreasing their bioavailability [55]. An experimental study found that the ferric iron-reducing antioxidant power value, an indicator of antioxidant activity, decreased as milk was added to tea [56]. The reduced antioxidant capacity of flavonoids after binding to proteins such as casein in milk may have attenuated the association between tea consumption and mortality. To understand more clearly the effects of adding milk to tea on the association between tea consumption and mortality, future prospective cohort studies that include information on how tea is consumed would need to be performed.

Because of the antioxidant properties of the polyphenols contained in tea, drinking tea has generally been considered a healthy dietary habit [57]. Antioxidative ability is an important factor in preventing disease and lowering the risk of premature death because oxidative stress is known to cause chronic disease by damaging normal cells and DNA. Tea flavonoids such as catechins or theaflavins have been found to decrease the oxidation of low-density lipoproteins, which cause atherosclerosis and increase the risk of CVD [58]. In addition, epigallocatechin-3-gallate (EGCG) has been shown to have preventive effects against cancer by inhibiting the activity of urokinase, which invades cells and forms metastases in human cancers [59], or by inducing apoptosis and cell cycle arrest specific to cancer cells [60].

This study has offered a comprehensive meta-analysis of prospective cohort studies to examine the associations between tea consumption and the risks of mortality from all causes, CVD, and cancer. Prospective cohort studies have relatively less recall or selection bias than other observational study designs. The present meta-analysis included a large number of people (1,956,549 total participants), including recently published large cohorts, and thus, the results could have considerably more statistical power. To identify the appropriate tea consumption level for minimizing mortality risk, we performed a dose-response analysis that could show changes in ESs as tea intake increased. Additionally, unlike previous meta-analyses on the

association between tea consumption and mortality, we excluded studies where all participants were patients suffering from a specific disease to examine the effect of tea consumption on mortality risk in the general population. Despite these strengths, several limitations should be mentioned as well. First, in most of the included studies, the data on tea consumption were assessed through a food frequency questionnaire, which allowed the possibility of misclassification. Fortunately, such misclassification of dietary intake tends to lead to a null result, so it is unlikely that the associations between tea consumption and mortality risk were exaggerated. Second, the volume of tea consumption was usually reported in cups, but cup size may differ from study to study. Third, the chemical composition of each cup of tea, such as its polyphenols, could vary by season, climate, and plant age, which may affect the associations between tea consumption and the risk of mortality [53]. Finally, the adjustment factors differed across the studies, which might have affected the pooled estimates in the meta-analysis. Most included studies controlled for potential confounders including smoking, alcohol consumption, and BMI, but the possibility of additional confounding factors cannot be completely ruled out.

In conclusion, the current systematic review and meta-analysis of prospective cohort studies suggested that moderate daily tea consumption (1.5–2 cups/day) was associated with lower risks of mortality from all causes, CVD, and cancer in the general population. Future well-designed, large-scale prospective cohort studies analyzing different types of tea are needed to develop dietary guidelines for tea consumption to reduce the risk of chronic disease and premature death.

Conflict of Interest: The authors declare no potential conflicts of interest.

Funding

This work was supported by the National Research Foundation of Korea, funded by the

Ministry of Science, ICT, and Future Planning (grant number NRF-2021R1F1A1050847).

The funders had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Author contributions

The authors' responsibilities were as follows— YK and YJ: study concept and design; YK: data collection and statistical analysis; YK: writing—original draft; YJ: writing—review and editing; YJ: study supervision; YK and YJ: interpretation of the data, critical revision of the paper for important intellectual content, and approval of the final paper for submission; and all authors: read and approved the final manuscript.

References

1. McKay DL, Blumberg JB. The role of tea in human health: an update. *J Am Coll Nutr.* 2002;21:1-13. <https://doi.org/10.1080/07315724.2002.10719187>.
2. He HF, Wei K, Yin JF, Ye Y. Insight into Tea Flavonoids: Composition and Chemistry. *Food Rev Int.* 2021;37:812-23. <https://doi.org/10.1080/87559129.2020.1721530>.
3. Agati G, Azzarello E, Pollastri S, Tattini M. Flavonoids as antioxidants in plants: location and functional significance. *Plant Sci.* 2012;196:67-76. <https://doi.org/10.1016/j.plantsci.2012.07.014>.
4. Maleki SJ, Crespo JF, Cabanillas B. Anti-inflammatory effects of flavonoids. *Food Chem.* 2019;299:125124. <https://doi.org/10.1016/j.foodchem.2019.125124>.
5. Kim Y, Je Y. Flavonoid intake and mortality from cardiovascular disease and all causes: A meta-analysis of prospective cohort studies. *Clin Nutr ESPEN.* 2017;20:68-77. <https://doi.org/10.1016/j.clnesp.2017.03.004>.
6. Kahn HA, Phillips RL, Snowdon DA, Choi W. Association between reported diet and all-cause mortality. Twenty-one-year follow-up on 27,530 adult Seventh-Day Adventists. *American journal of epidemiology.* 1984;119:775-87. <https://doi.org/10.1093/oxfordjournals.aje.a113798>.
7. Hertog MG, Feskens EJ, Hollman PC, Katan MB, Kromhout D. Dietary antioxidant flavonoids and risk of coronary heart disease: the Zutphen Elderly Study. *Lancet.* 1993;342:1007-11. [https://doi.org/10.1016/0140-6736\(93\)92876-u](https://doi.org/10.1016/0140-6736(93)92876-u).
8. Klatsky AL, Armstrong MA, Friedman GD. Coffee, tea, and mortality. *Ann Epidemiol.* 1993;3:375-81. [https://doi.org/10.1016/1047-2797\(93\)90064-b](https://doi.org/10.1016/1047-2797(93)90064-b).
9. Hertog MG, Sweetnam PM, Fehily AM, Elwood PC, Kromhout D. Antioxidant flavonols and ischemic heart disease in a Welsh population of men: the Caerphilly Study. *Am J Clin Nutr.* 1997;65:1489-94. <https://doi.org/10.1093/ajcn/65.5.1489>.
10. Woodward M, Tunstall-Pedoe H. Coffee and tea consumption in the Scottish Heart Health Study follow up: conflicting relations with coronary risk factors, coronary disease, and all cause mortality. *J Epidemiol Community Health.* 1999;53:481-7. <https://doi.org/10.1136/jech.53.8.481>.
11. Nakachi K, Matsuyama S, Miyake S, Suganuma M, Imai K. Preventive effects of drinking green tea on cancer and cardiovascular disease: epidemiological evidence for multiple targeting prevention. *Biofactors.* 2000;13:49-54. <https://doi.org/10.1002/biof.5520130109>.
12. Hirvonen T, Pietinen P, Virtanen M, Ovaskainen ML, Häkkinen S, Albanes D, et al. Intake of flavonols and flavones and risk of coronary heart disease in male smokers. *Epidemiology.* 2001;12:62-7. <https://doi.org/10.1097/00001648-200101000-00011>.
13. Iwai N, Ohshiro H, Kurozawa Y, Hosoda T, Morita H, Funakawa K, et al. Relationship between coffee and green tea consumption and all-cause mortality in a cohort of a rural Japanese population. *J Epidemiol.* 2002;12:191-8. <https://doi.org/10.2188/jea.12.191>.
14. Khan MM, Goto R, Kobayashi K, Suzumura S, Nagata Y, Sonoda T, et al. Dietary habits and cancer mortality among middle aged and older Japanese living in hokkaido, Japan by cancer site and sex. *Asian Pac J Cancer Prev.* 2004;5:58-65.
15. Andersen LF, Jacobs DR, Jr., Carlsen MH, Blomhoff R. Consumption of coffee is associated with reduced risk of death attributed to inflammatory and cardiovascular diseases in the Iowa Women's Health Study. *Am J Clin Nutr.* 2006;83:1039-46. <https://doi.org/10.1093/ajcn/83.5.1039>.
16. Paganini-Hill A, Kawas CH, Corrada MM. Non-alcoholic beverage and caffeine consumption and mortality: the Leisure World Cohort Study. *Prev Med.* 2007;44:305-10. <https://doi.org/10.1016/j.ypmed.2006.12.011>.
17. Suzuki E, Yorifuji T, Takao S, Komatsu H, Sugiyama M, Ohta T, et al. Green tea consumption and mortality among Japanese elderly people: the prospective Shizuoka elderly cohort. *Ann Epidemiol.* 2009;19:732-9. <https://doi.org/10.1016/j.annepidem.2009.06.003>.
18. de Koning Gans JM, Uiterwaal CS, van der Schouw YT, Boer JM, Grobbee DE, Verschuren WM, et al. Tea and coffee consumption and cardiovascular morbidity and mortality. *Arterioscler Thromb Vasc Biol.* 2010;30:1665-71. <https://doi.org/10.1161/atvbaha.109.201939>.
19. Mineharu Y, Koizumi A, Wada Y, Iso H, Watanabe Y, Date C, et al. Coffee, green tea, black tea and oolong tea consumption and risk of mortality from cardiovascular disease in Japanese men and women. *J Epidemiol Community Health.* 2011;65:230-40. <https://doi.org/10.1136/jech.2009.097311>.
20. Bertoia ML, Triche EW, Michaud DS, Baylin A, Hogan JW, Neuhaus ML, et al. Long-term alcohol and caffeine intake and risk of sudden cardiac death in women. *Am J Clin Nutr.* 2013;97:1356-63. <https://doi.org/10.3945/ajcn.112.044248>.
21. Gardener H, Rundek T, Wright CB, Elkind MS, Sacco RL. Coffee and tea consumption are inversely associated with mortality in a multiethnic urban population. *J Nutr.* 2013;143:1299-308. <https://doi.org/10.3945/jn.112.173807>.
22. Liu J, Liu S, Zhou H, Hanson T, Yang L, Chen Z, et al. Association of green tea consumption with mortality

- from all-cause, cardiovascular disease and cancer in a Chinese cohort of 165,000 adult men. *Eur J Epidemiol.* 2016;31:853-65. <https://doi.org/10.1007/s10654-016-0173-3>.
23. Ivey KL, Jensen MK, Hodgson JM, Eliassen AH, Cassidy A, Rimm EB. Association of flavonoid-rich foods and flavonoids with risk of all-cause mortality. *Br J Nutr.* 2017;117:1470-7. <https://doi.org/10.1017/s0007114517001325>.
 24. Lim WH, Wong G, Lewis JR, Lok CE, Polkinghorne KR, Hodgson J, et al. Total volume and composition of fluid intake and mortality in older women: a cohort study. *BMJ Open.* 2017;7:e011720. <https://doi.org/10.1136/bmjopen-2016-011720>.
 25. Yan Y, Sui X, Yao B, Lavie CJ, Blair SN. Is There a Dose-Response Relationship between Tea Consumption and All-Cause, CVD, and Cancer Mortality? *J Am Coll Nutr.* 2017;36:281-6. <https://doi.org/10.1080/07315724.2016.1261054>.
 26. van den Brandt PA. Coffee or Tea? A prospective cohort study on the associations of coffee and tea intake with overall and cause-specific mortality in men versus women. *Eur J Epidemiol.* 2018;33:183-200. <https://doi.org/10.1007/s10654-018-0359-y>.
 27. Wang X, Liu F, Li J, Yang X, Chen J, Cao J, et al. Tea consumption and the risk of atherosclerotic cardiovascular disease and all-cause mortality: The China-PAR project. *Eur J Prev Cardiol.* 2020;27:1956-63. <https://doi.org/10.1177/2047487319894685>.
 28. Chen Y, Zhang Y, Zhang M, Yang H, Wang Y. Consumption of coffee and tea with all-cause and cause-specific mortality: a prospective cohort study. *BMC Med.* 2022;20:449. <https://doi.org/10.1186/s12916-022-02636-2>.
 29. Inoue-Choi M, Ramirez Y, Cornelis MC, Berrington de González A, Freedman ND, Loftfield E. Tea Consumption and All-Cause and Cause-Specific Mortality in the UK Biobank : A Prospective Cohort Study. *Ann Intern Med.* 2022;175:1201-11. <https://doi.org/10.7326/m22-0041>.
 30. Shin S, Lee JE, Loftfield E, Shu XO, Abe SK, Rahman MS, et al. Coffee and tea consumption and mortality from all causes, cardiovascular disease and cancer: a pooled analysis of prospective studies from the Asia Cohort Consortium. *Int J Epidemiol.* 2022;51:626-40. <https://doi.org/10.1093/ije/dyab161>.
 31. Qiu W, Cai A, Xiao X, Xia S, Li L, Feng Y. Alcohol intake masked the protective effects of tea consumption against all-cause mortality and blood pressure progression: Findings from CHNS cohort, 1993-2011. *Nutrition.* 2023;111:112003. <https://doi.org/10.1016/j.nut.2023.112003>.
 32. Teramoto M, Muraki I, Yamagishi K, Tamakoshi A, Iso H. Green Tea and Coffee Consumption and All-Cause Mortality Among Persons With and Without Stroke or Myocardial Infarction. *Stroke.* 2021;52:957-65. <https://doi.org/10.1161/strokeaha.120.032273>.
 33. Tang J, Zheng JS, Fang L, Jin Y, Cai W, Li D. Tea consumption and mortality of all cancers, CVD and all causes: a meta-analysis of eighteen prospective cohort studies. *Br J Nutr.* 2015;114:673-83. <https://doi.org/10.1017/s0007114515002329>.
 34. Zhang C, Qin YY, Wei X, Yu FF, Zhou YH, He J. Tea consumption and risk of cardiovascular outcomes and total mortality: a systematic review and meta-analysis of prospective observational studies. *Eur J Epidemiol.* 2015;30:103-13. <https://doi.org/10.1007/s10654-014-9960-x>.
 35. Chung M, Zhao N, Wang D, Shams-White M, Karlsen M, Cassidy A, et al. Dose-Response Relation between Tea Consumption and Risk of Cardiovascular Disease and All-Cause Mortality: A Systematic Review and Meta-Analysis of Population-Based Studies. *Adv Nutr.* 2020;11:790-814. <https://doi.org/10.1093/advances/nmaa010>.
 36. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71. <https://doi.org/10.1136/bmj.n71>.
 37. Wells GA SB, O'Connell D. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analysis. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp.
 38. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Controlled clinical trials.* 1986;7:177-88.
 39. Cochran WG. The combination of estimates from different experiments. *Biometrics.* 1954;10:101-29.
 40. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ.* 2003;327:557-60. <https://doi.org/10.1136/bmj.327.7414.557>.
 41. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics.* 1994;50:1088-101.
 42. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ.* 1997;315:629-34.
 43. Greenland S, Longnecker MP. Methods for trend estimation from summarized dose-response data, with applications to meta-analysis. *American journal of epidemiology.* 1992;135:1301-9.
 44. Berlin JA, Longnecker MP, Greenland S. Meta-analysis of epidemiologic dose-response data. *Epidemiology.* 1993;4:218-28.
 45. Orsini N, Bellocco R, Greenland S. Generalized least squares for trend estimation of summarized dose-response data. *Stata J.* 2006;6:40-57.
 46. Orsini N, Li R, Wolk A, Khudyakov P, Spiegelman D. Meta-analysis for linear and nonlinear dose-response

- relations: examples, an evaluation of approximations, and software. *American journal of epidemiology*. 2012;175:66-73. <https://doi.org/10.1093/aje/kwr265>.
47. Li X, Zeng J, Chen B, Yan Q, Cui Y, Xu W, et al. Daily higher tea consumption is associated with a reduced risk of type 2 diabetes: A cohort study and updated systematic review and meta-analysis. *Nutr Res*. 2023;118:116-27. <https://doi.org/10.1016/j.nutres.2023.08.002>.
48. Wang ZM, Zhao D, Wang H, Wang QM, Zhou B, Wang LS. Green tea consumption and the risk of coronary heart disease: A systematic review and meta-analysis of cohort studies. *Nutrition Metabolism and Cardiovascular Diseases*. 2023;33:715-23. <https://doi.org/10.1016/j.numecd.2023.01.017>.
49. Wang ZM, Chen B, Zhou B, Zhao D, Wang LS. Green tea consumption and the risk of stroke: A systematic review and meta-analysis of cohort studies. *Nutrition*. 2023;107:111936. <https://doi.org/10.1016/j.nut.2022.111936>.
50. Zheng F, Chen K, Zhong J, Tang S, Xu S, Lu W, et al. Association between Different Types of Tea Consumption and Risk of Gynecologic Cancer: A Meta-Analysis of Cohort Studies. *Nutrients*. 2023;15. <https://doi.org/10.3390/nu15020403>.
51. Huang YQ, Lu X, Min H, Wu QQ, Shi XT, Bian KQ, et al. Green tea and liver cancer risk: A meta-analysis of prospective cohort studies in Asian populations. *Nutrition*. 2016;32:3-8. <https://doi.org/10.1016/j.nut.2015.05.021>.
52. Zhao LG, Li ZY, Feng GS, Ji XW, Tan YT, Li HL, et al. Tea Drinking and Risk of Cancer Incidence: A Meta-Analysis of Prospective Cohort Studies and Evidence Evaluation. *Adv Nutr*. 2021;12:402-12. <https://doi.org/10.1093/advances/nmaa117>.
53. Mukhtar H, Ahmad N. Tea polyphenols: prevention of cancer and optimizing health. *Am J Clin Nutr*. 2000;71:1698S-702S; discussion 703S-4S. <https://doi.org/10.1093/ajcn/71.6.1698S>.
54. Papadopoulou A, Frazier RA. Characterization of protein-polyphenol interactions. *Trends in Food Science & Technology*. 2004;15:186-90. <https://doi.org/10.1016/j.tifs.2003.09.017>.
55. Arts MJ, Haenen GR, Wilms LC, Beetstra SA, Heijnen CG, Voss HP, et al. Interactions between flavonoids and proteins: effect on the total antioxidant capacity. *J Agric Food Chem*. 2002;50:1184-7. <https://doi.org/10.1021/jf010855a>.
56. Ryan L, Petit S. Addition of whole, semiskimmed, and skimmed bovine milk reduces the total antioxidant capacity of black tea. *Nutr Res*. 2010;30:14-20. <https://doi.org/10.1016/j.nutres.2009.11.005>.
57. Rietveld A, Wiseman S. Antioxidant effects of tea: evidence from human clinical trials. *J Nutr*. 2003;133:3285S-92S. <https://doi.org/10.1093/jn/133.10.3285S>.
58. Ishikawa T, Suzukawa M, Ito T, Yoshida H, Ayaori M, Nishiwaki M, et al. Effect of tea flavonoid supplementation on the susceptibility of low-density lipoprotein to oxidative modification. *Am J Clin Nutr*. 1997;66:261-6. <https://doi.org/10.1093/ajcn/66.2.261>.
59. Jankun J, Selman SH, Swiercz R, Skrzypczak-Jankun E. Why drinking green tea could prevent cancer. *Nature*. 1997;387:561. <https://doi.org/10.1038/42381>.
60. Ahmad N, Feyes DK, Nieminen AL, Agarwal R, Mukhtar H. Green tea constituent epigallocatechin-3-gallate and induction of apoptosis and cell cycle arrest in human carcinoma cells. *J Natl Cancer Inst*. 1997;89:1881-6. <https://doi.org/10.1093/jnci/89.24.1881>.

Figure Legends

Figure 1. Flow chart of study selection.

Figure 2. Forest plots of the prospective cohort studies of all-cause mortality for high versus low tea consumption.

Figure 3. Pooled dose-response association between tea consumption and all-cause mortality. Solid lines represent effect estimates (ESs); dashed lines represent 95% confidence intervals.

Figure 4. Forest plots of the prospective cohort studies of cardiovascular disease mortality for high *versus* low tea consumption.

Figure 5. Pooled dose-response association between tea consumption and cardiovascular disease mortality. Solid lines represent effect estimates (ESs); dashed lines represent 95% confidence intervals.

Figure 6. Forest plots of the prospective cohort studies of cancer mortality for high *versus* low tea consumption.

Figure 7. Pooled dose-response association between tea consumption and cancer mortality. Solid lines represent effect estimates (ESs); dashed lines represent 95% confidence intervals.

Epub ahead of print

Table 1. Characteristics of studies included in the meta-analysis ~~on~~of tea consumption and mortality from all -causes, ~~cardiovascular disease (CVD)~~, and cancer

First author, year	Country	Cohort name	Follow-up period (years)	Age at baseline	Study size		Sex	Tea consumption	Cause of death	Adjustment for covariates
					Subjects	No. of deaths				
Kahn, 1984 [6]	USA	Seventh-Day Adventists	21 y	≥30 y	21,022	5,679	Men and women	<1 vs. ≥1 cups/day	All causes	Age, gender <u>sex</u> , smoking history, history of heart disease, stroke, hypertension, diabetes, or cancer; age at initial exposure to the Adventist Church
Hertog, 1993 [7]	Netherlands	Zutphen Elderly Study	5 y	65-84 y	805	43	Men	0-250 vs. >500 mL/day	Coronary heart disease (CHD)	Age, intake of total energy, saturated fatty acids, cholesterol, alcohol, coffee, vitamin C, vitamin E, beta-carotene, dietary fiber, history of myocardial infarction, BMI, smoking, serum total and HDL cholesterol, systolic blood pressure
Klatsky, 1993 [8]	USA	Northern California Kaiser Permanente Medical Care Program	8 y	NA	125,356	4,208	Men and women	0 vs. ≥4 cups/day	All causes	Age, sex, BMI, smoking, alcohol, race, education, marital status
Hertog, 1997 [9]	UK	Caerphilly study	14 y	45-59y	1,900	338 ^c 104 ^d	Men	0-300 vs. >1200 mL/day	All causes Cancer	Age, smoking, social class, BMI, intakes of total energy, alcohol, fat, vitamin C, vitamin E, and β-carotene
Woodward, 1999 [10]	Scotland	Scottish Heart Health Study	7.7 y	40-59 y	11,507	588 ^c 206 ^e	Men and women	0 vs. ≥5 cups/day	All causes CHD	Age, housing tenure, activity at work, activity in leisure, cigarette smoking status, BMI, Bortner score, cotinine, systolic blood pressure, fibrinogen, total cholesterol, HDL-cholesterol, triglycerides, alcohol, vitamin C, and coffee

Nakachi, 2000 [11]	Japan	Saitama Prefecture	11 y	>40 y	8,552	222	Men and women	≤3 cups/day vs. ≥10	CVD	Age, cigarette smoking, alcohol consumption, intake of meat, and relative body weight
Hirvonen, 2001 [12]	Finland	Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study	6.1 y	50-69 y	25,372	815	Men	<1 vs. ≥1 cups/day	CVD	Age, supplementation group, systolic and diastolic blood pressure, serum total cholesterol, serum high-density lipoprotein HDL cholesterol, BMI, smoking years, number of cigarettes smoked daily, history of diabetes mellitus and coronary heart disease CHD, marital status, education, leisure-time physical activity
Iwai, 2002 [13]	Japan	Tottori Prefecture	9.9 y	40-79 y	2,855	361 ^c 61 ^d	Men and women	<0.5 cups/day vs. ≥4	All causes Cancer	Age, smoking, alcohol, history of selected diseases, physical activity, educational status
Khan, 2004 [14]	Japan	Hokkaido Prefecture	13.8 y	≥40 y	3,158	244	Men and women	Never + several times/year + several times/month vs. several times/week + everyday daily	Cancer	Age, — health education, health examination, health status, smoking-
Andersen, 2006 [15]	USA	Iowa Woman's Health Study	15 y	55-69 y	27,312	4,265 ^c 1,411 ^f	Women	0 vs. >3 cups/day	All causes CVD	Age, smoking, intake of alcohol, BMI, waist-hip ratio, education, physical activity, use of estrogens, use of multivitamin supplements, energy intake, and intakes of whole and refined grain, red meat, fish, seafood, total fruit and vegetables
Paganini-Hill, 2007 [16]	USA	Leisure World Cohort Study	23 y	≥44 y	13,624	11,386	Men and women	0 vs. ≥2 cups/day	All causes	Age, sex, smoking, exercise, BMI, alcohol intake, and histories of hypertension, angina, heart attack, stroke, diabetes, rheumatoid arthritis, and

cancer

Suzuki, 2009 [17]	Japan	Shizuoka Elderly Cohort	6 y	65-84 y	12,251	1,224 ^c 405 ^f 400 ^d	Men and women	<1 vs. ≥7 cups/day	All causes CVD Cancer
Koning Gans, 2010 [18]	Netherlands	European Prospective Investigation into Cancer and Nutrition-Netherlands (EPIC-NL), MORGEN	13 y	20-69 y	37,514	1,405 ^c 123 ^f	Men and women	<1 vs. >6 cups/day	All causes CVD
Mineharu, 2011 [19]	Japan	Japan Collaborative Cohort Study for Evaluation of Cancer Risk (JACC)	13.1 y	40-79 y	82,655	3,125	Men and women	<1 cup/week vs. ≥6 cups/day	CVD
Bertoia, 2013 [20]	USA	Women's Health Initiative Observational Study (WHI)	11 y	50-79 y	92,847	205	Women	0 vs. ≥4 cups/day	CVD
Gardner, 2013 [21]	USA	Northern Manhattan Study	11 y	>40 y	2,461	863 ^c 342 ^f 160 ^d	Men and women	1 cup/month vs. ≥2 cups/day	All causes CVD Cancer

Age, sex, smoking status, alcohol consumption, BMI, ~~the~~ frequency of physical activity

Age, sex, ~~cohort~~ (strata), educational, physical activity, smoking status, waist circumference, menopausal status, alcohol, coffee, vitamin C, level, fiber, consumption, energy, saturated fat:

Age, BMI, smoking status, alcohol intakes, history of hypertension, history of diabetes, education, waking hours, hours of sports participation, perceived mental stress, multivitamin use, vitamin E supplement use; ~~total~~ consumption of ~~total~~ fruits, ~~total~~ vegetables, ~~total~~ beans, ~~total~~ meat, and ~~total~~ fish; and total daily energy intake

Age, total energy intake, race, income, smoking status, physical activity, waist-to-hip ratio, BMI, atrial fibrillation, coronary artery disease, heart failure, diabetes, high cholesterol, hypertension, pulse in 60 s, and hormone use

Age, ~~gender~~ sex, BMI, race, education, pack-years of smoking, alcohol consumption, energy, protein, carbohydrates, total fat, saturated fat, history of vascular risk factors, other

										non-water beverage consumption, coffee additives (milk, cream, nondairy creamer), coffee
Liu, 2016 [22]	China	Chinese Prospective Smoking Study	11 y	>40 y	164,681	32,700	Men	0 vs. >10 g/day	All causes CVD Cancer	Age, BMI, marital status, urban locality, education, job status, smoking status, alcohol drinking, times of weekly consumption for fish, meat, poultry consumption, egg, and milk, black tea drinker, jasmine tea drinker, other tea drinker
Ivey, 2017 [23]	USA	Nurses' Health Study II	18 y	25-42 y	93,145	1,894 ^c 189 ^f 887 ^d	Women	0 vs. >1 times/week	All causes CVD Cancer	Age, BMI, smoking status, menopausal status, family history (of diabetes, cancer, and myocardial infarction), multivitamin supplement use, aspirin use, race, type 2 diabetes, hypercholesterolemia, hypertension, physical activity, energy intake, alcohol consumption, and the Alternative Health Eating Index (minus alcohol) score
Lim, 2017 [24]	Australia	Calcium Intake Fracture Outcome Study	10 y	≥70 y	1,055	362 ^c 142 ^f	Women	1 _____ cup/day increment	All causes CVD	Age, smoking history, SES, diabetes status, hypertension, blood pressure, prevalent CVD, medications and treatment code (calcium supplementation vs. no calcium supplementation), fluid status, BMI, estimated glomerular filtration rate
Yan, 2017 [25]	USA	Aerobics Center Longitudinal Study	16 y	20-82 y	11,808	842 ^c 250 ^f 345 ^d	Men and women	0 vs. >14 cups/week	All causes CVD Cancer	Age, sex, baseline examination year, regular coffee use, decaffeinated coffee use, herbal tea use,

Brandt, 2018 [26]	Netherlands	Netherlands Cohort Study (NLCS)	10 y	55–69 y	120,852	8,665 ^c 2,927 ^f 3,849 ^d	Men and women	0 vs. ≥5 cups/day	All causes CVD Cancer	physical inactivity, BMI, smoking, alcohol consumption, metabolic equivalents Age—, cigarette smoking status, number of cigarettes smoked per day, and years of smoking, history of physician-diagnosed hypertension and diabetes, body height, —BMI, non-occupational physical activity, highest level of education, intake of alcohol, nuts, vegetables and fruit, tea, energy, use of nutritional supplements, postmenopausal hormone replacement therapy (in women)
Wang, 2020 [27]	China	China-PAR project (3 cohorts) ^a	7.3 y	52y, mean	100,902	5,479 ^c 1,477 ^f	Men and women	<3 vs. ≥3 times/week	All causes CVD	Age, sex, region, <u>residential</u> area, cohort, educational level, family history of atherosclerotic cardiovascular disease CVD, smoking, drinking, physical activity level, dietary factors, BMI, systolic blood pressure, fasting blood glucose, total cholesterol, <u>high density lipoprotein</u> HDL cholesterol
Teramoto, 2021 [32]	Japan	Japan Collaborative Cohort Study for Evaluation of Cancer Risk (JACC)	18.5 y	40–79 y	44,521	8,666	Men and women	0 vs. ≥7 cups/day	All causes	Age, sex, coffee consumption, history of hypertension, history of diabetes, BMI, smoking status, alcohol consumption, hours of exercise, hours of walking, perceived mental stress, educational level, regular employment, dietary intakes of vegetable, fish, fruits, and soybeans

Chen, 2022 [28]	UK	UK Biobank	12.1 y	37–73 y	498,158	34,699 ^c 6,663 ^f	Men and women	0 vs. ≥5 cups/day	All causes CVD	Age, sex, —ethnicity, educational levels, BMI, smoking status, alcohol intake frequency, physical activity, dietary pattern, general health status, hypertension, diabetes, depression, coffee consumption
Inoue-Choi, 2022 [29]	UK	UK Biobank	11.2 y	40–69 y	498,043	15,790	Men and women	0 vs. ≥10 cups/day	Cancer	Age, sex, race and ethnicity, Townsend deprivation score, general health status, cancer, cardiovascular disease CVD, diabetes—, BMI, tobacco smoking, physical activity, alcohol intake, coffee intake, dietary intake including vegetables, fruits, red meat, and processed meat, assessment centers
Shin, 2022 [30]	Asia	Asia Cohort Consortium (12 cohorts) ^b	6.5 -22.7 y	54.3 y, mean	528,504	94,744 ^c 22,850 ^f 27,207 ^d	Men and women	0 vs. ≥5 cups/day	All causes CVD Cancer	Age, BMI, smoking status, alcohol intake, educational level, energy intake, coffee consumption
Qiu, 2023 [31]	China	China Health and Nutrition Survey (CHNS)	17.9 y	54.4 y, mean	6,387	580	Men and women	0 vs. 3-4 cups/day	All causes	Age, sex, marital status, educational level, nationality, residential area, smoking status, occupation, level of annual income, mean systolic blood pressure, mean BMI, mean waist and hip circumferences, hypertension, diabetes mellitus, myocardial infarction, stroke, malignant tumor

^a China 3 cohort is China Multi-Center Collaborative Study of Cardiovascular Epidemiology, International Collaborative Study of CVD in Asia, and Community Intervention of Metabolic Syndrome in China & Chinese Family Health Study

^b Asia Cohort Consortium includes Japan Public Health Center-based Prospective Study (JPHC) I, II, Miyagi Cohort, Ohsaki National Health Insurance Cohort Study, Life Span Study Cohort (RERF), 3 Prefecture Miyagi, 3 Prefecture Aichi, —Seoul Male Cancer Cohort (Seoul Male), Korean Multi-center Cancer Cohort Study (KMCC), Shanghai Men's Health Study (SMHS), Shanghai Women's Health Study (SWHS), and Singapore Chinese Health Study (SCHS)

^c Death from all causes

^d Death from cancer

^e Death from CHD

^f Death from CVD

CVD, cardiovascular disease; CHD, coronary heart disease; BMI, body mass index; HDL, high-density lipoprotein; SES, socioeconomic status.

Epub ahead of print

Table 2. Summary of pooled effect estimates (ESs) of mortality from all-causes, CVD, and cancer by tea consumption

	No. of studies	ES	95% CI	P for difference
<i>All-cause mortality</i>				
High versus low tea consumption				
All studies	20	0.90	0.86-0.95	
Stratified by sex				
Men	8	0.87	0.77-0.98	0.97
Women	9	0.90	0.79-1.01	
Stratified by geographical region				
Asia	7	0.84	0.77-0.91	
US	7	0.92	0.83-1.02	0.32 ^a
Europe	5	1.12	0.88-1.42	0.05 ^a
Oceania	1	0.92	0.86-0.98	0.57 ^a
Stratified by follow-up years				
≥Median	10	0.96	0.88-1.05	0.17
<Median	10	0.86	0.81-0.92	
Stratified by number of subjects				
≥Median	10	0.89	0.85-0.94	0.87
<Median	10	0.91	0.80-1.04	
<i>CVD mortality</i>				
High versus low tea consumption				
All studies	18	0.86	0.79-0.94	
Stratified by sex				
Men	10	0.84	0.71-0.99	0.83
Women	11	0.86	0.75-0.99	
Stratified by geographical region				
Asia	6	0.75	0.65-0.88	
US	5	0.91	0.72-1.14	0.31 ^b
Europe	6	1.00	0.78-1.26	0.17 ^b
Oceania	1	0.90	0.79-0.94	0.55 ^b
Stratified by follow-up years				
≥Median	11	0.86	0.80-0.92	0.76
<Median	7	0.84	0.68-1.03	
Stratified by number of subjects				
≥Median	9	0.84	0.79-0.89	0.76
<Median	9	0.84	0.69-1.03	
<i>Cancer mortality</i>				
High versus low tea consumption				
All studies	11	0.90	0.78-1.03	
Stratified by sex				
Men	7	0.91	0.77-1.09	0.93
Women	5	0.93	0.70-1.23	
Stratified by geographical region				
Asia	5	0.92	0.74-1.15	

서식 있음: 글꼴: 기움임꼴 없음

서식 있음: 글꼴: 기움임꼴 없음

서식 있음: 글꼴: 기움임꼴 없음

서식 있음: 글꼴: 기움임꼴 없음

US	3	0.70	0.46-1.07	0.25 ^c
Europe	3	1.00	0.80-1.24	0.64 ^c
Stratified by follow-up years				
≥ Median	6	0.97	0.82-1.14	0.24
< Median	5	0.80	0.70-0.92	
Stratified by number of subjects				
≥ Median	6	0.88	0.76-1.03	0.72
< Median	5	0.94	0.65-1.35	

^a*P*-value difference in ESs of all-cause mortality for US *versus* Asia (*P*_E=0.32), Europe *versus* Asia (*P*_E=0.05), and Oceania *versus* Asia (*P*_E=0.57).

^b*P*-value difference in ESs of CVD mortality for US *versus* Asia (*P*_E=0.31), Europe *versus* Asia (*P*_E=0.17), and Oceania *versus* Asia (*P*_E=0.55).

^c*P*-value difference in ESs of cancer mortality for US *versus* Asia (*P*_E=0.25) and Europe *versus* Asia (*P*_E=0.64).

ES, effect estimate; CVD, cardiovascular disease.

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

서식 있음: 글꼴: 기울임꼴 없음

Figure 1

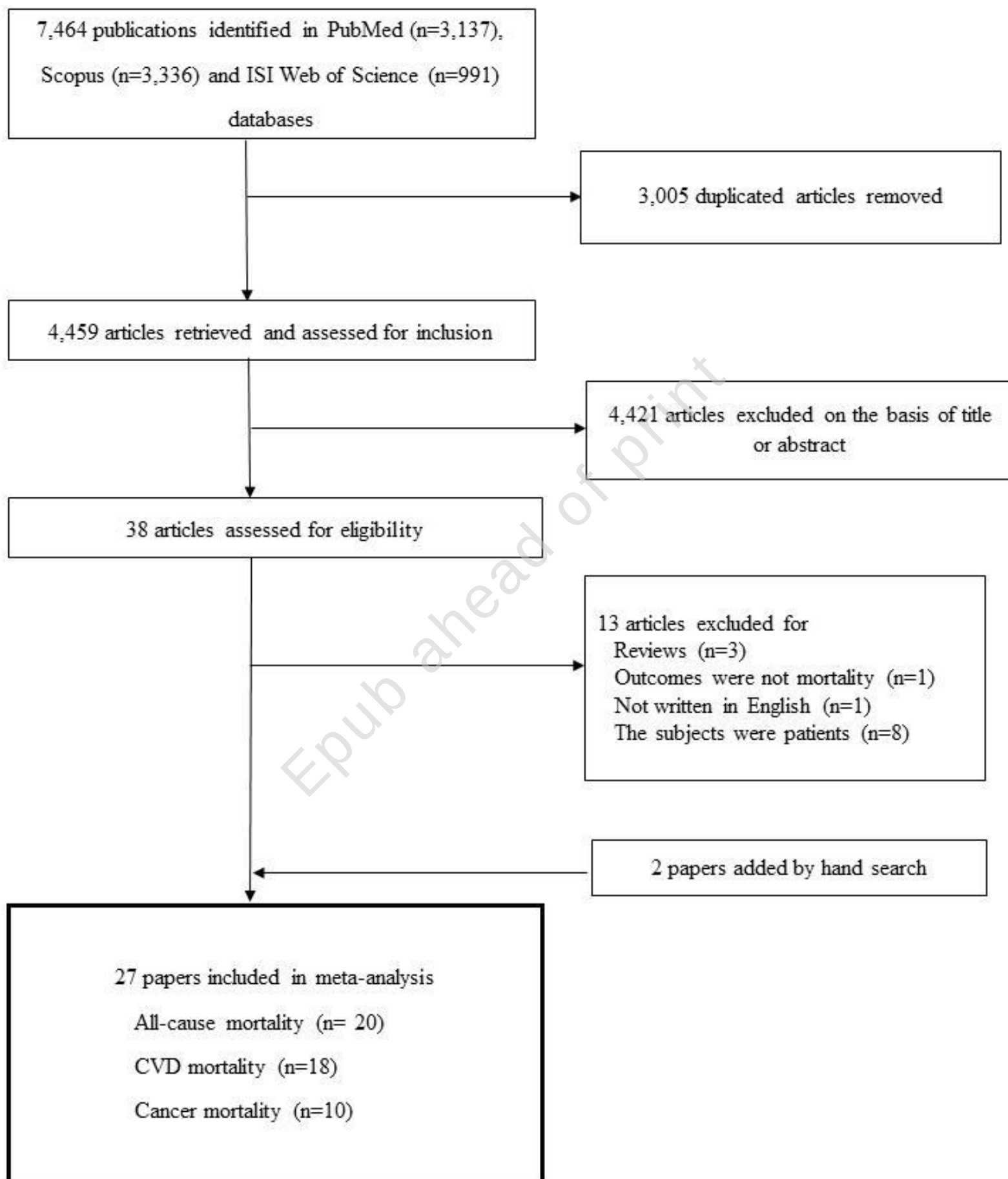


Figure 2

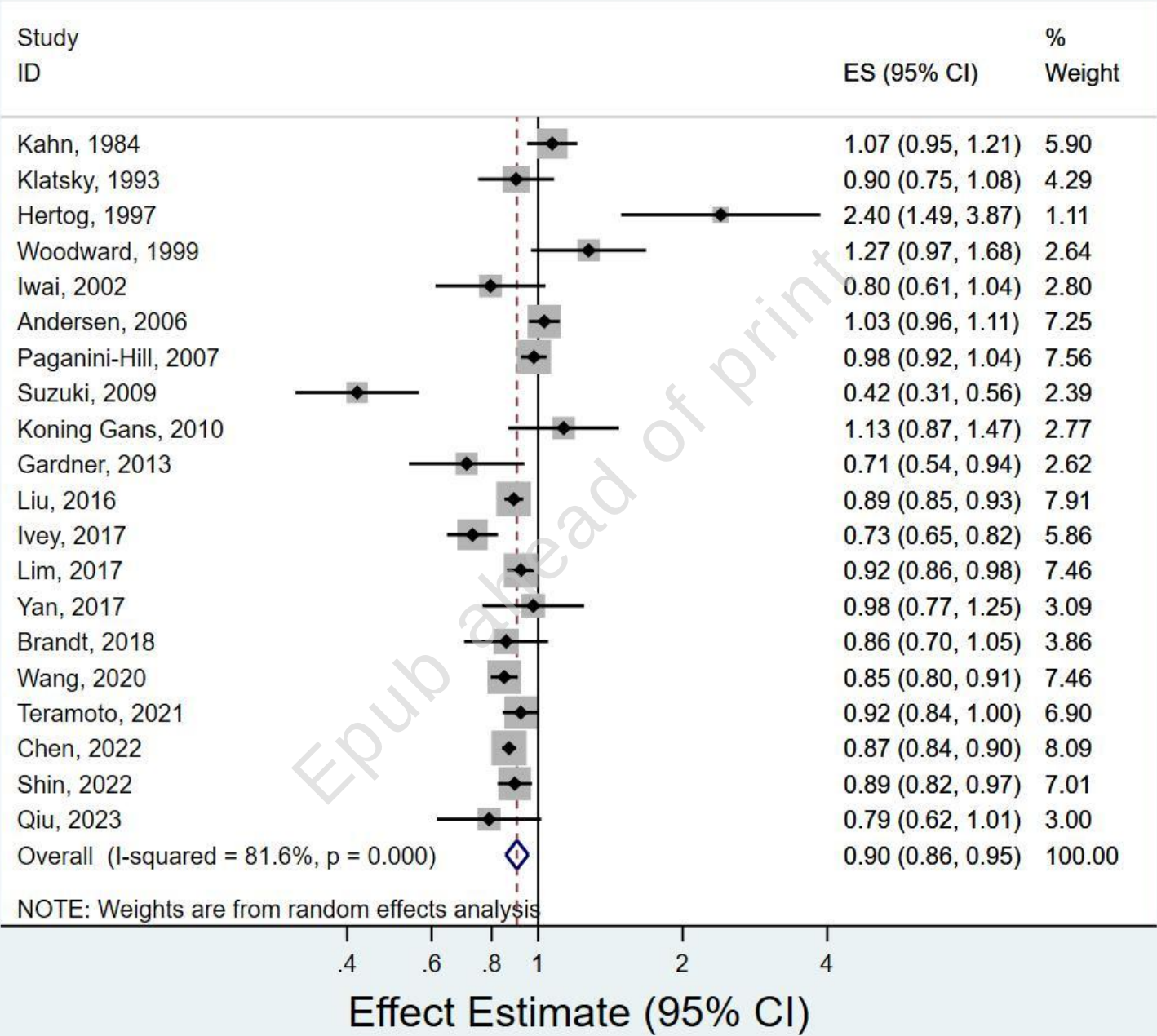
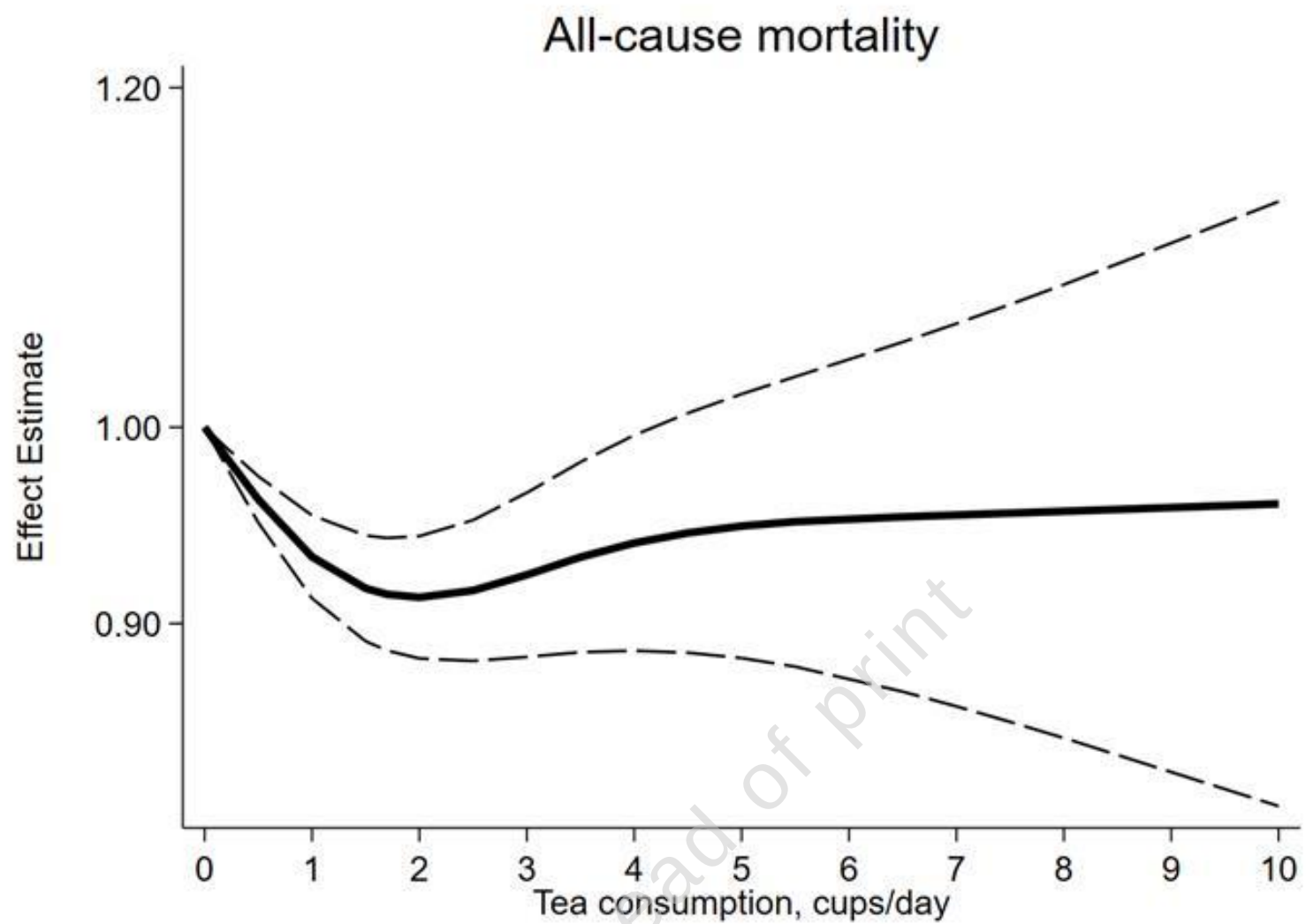


Figure 3



Cups/day	ES	95% CI
0	1.00 (reference)	
0.5	0.96	(0.95-0.97)
1	0.93	(0.91-0.95)
1.5	0.92	(0.89-0.94)
2	0.91	(0.88-0.94)
2.5	0.92	(0.88-0.95)
3	0.92	(0.88-0.97)
3.5	0.93	(0.89-0.98)
4	0.94	(0.89-1.00)
4.5	0.94	(0.89-1.01)
5	0.95	(0.88-1.02)
5.5	0.95	(0.88-1.03)
6.5	0.95	(0.87-1.05)
7	0.95	(0.86-1.06)
7.5	0.95	(0.85-1.07)
8	0.96	(0.85-1.08)
10	0.96	(0.82-1.13)
$P_{\text{nonlinearity}}$		<0.0001

Figure 4

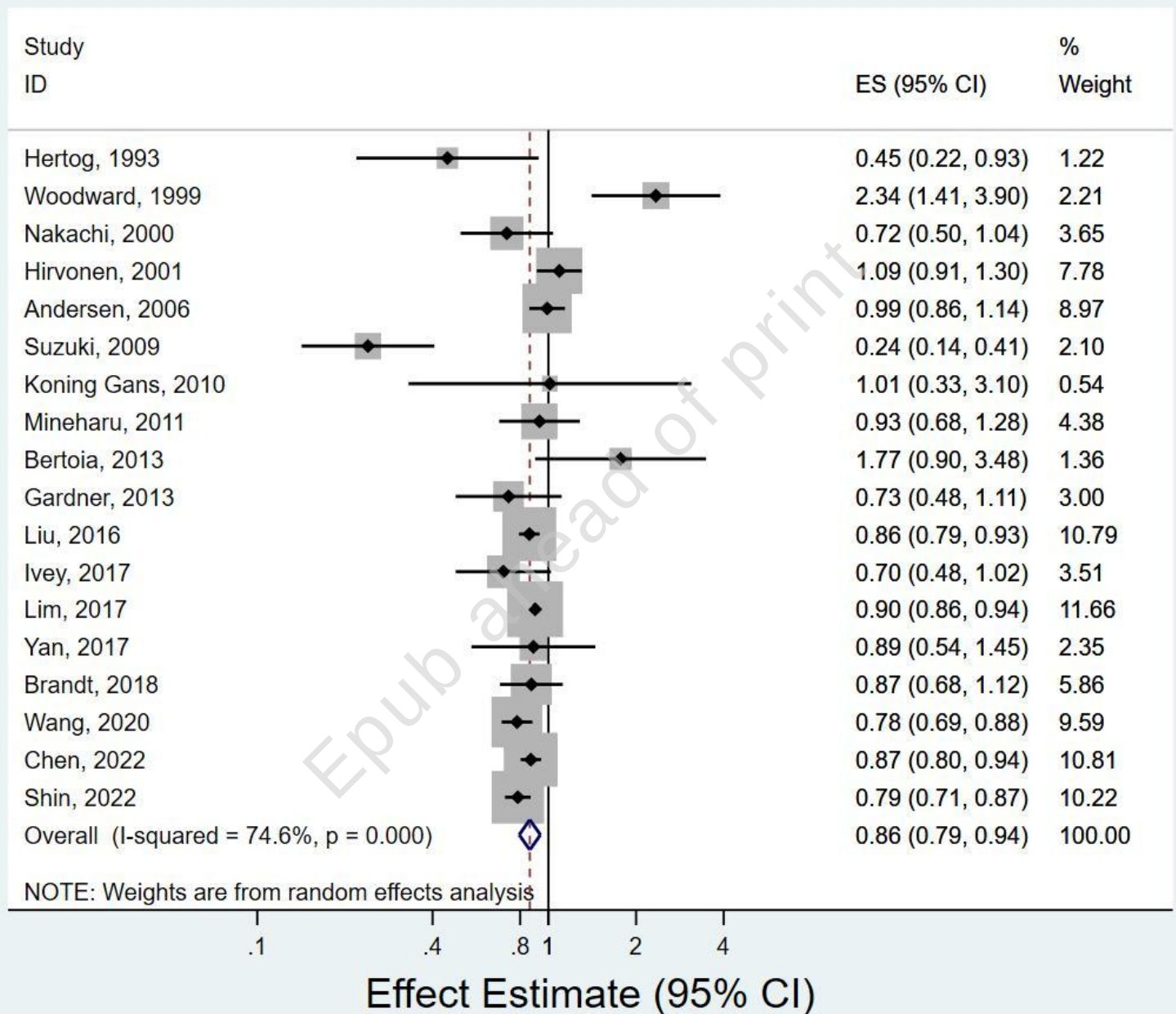
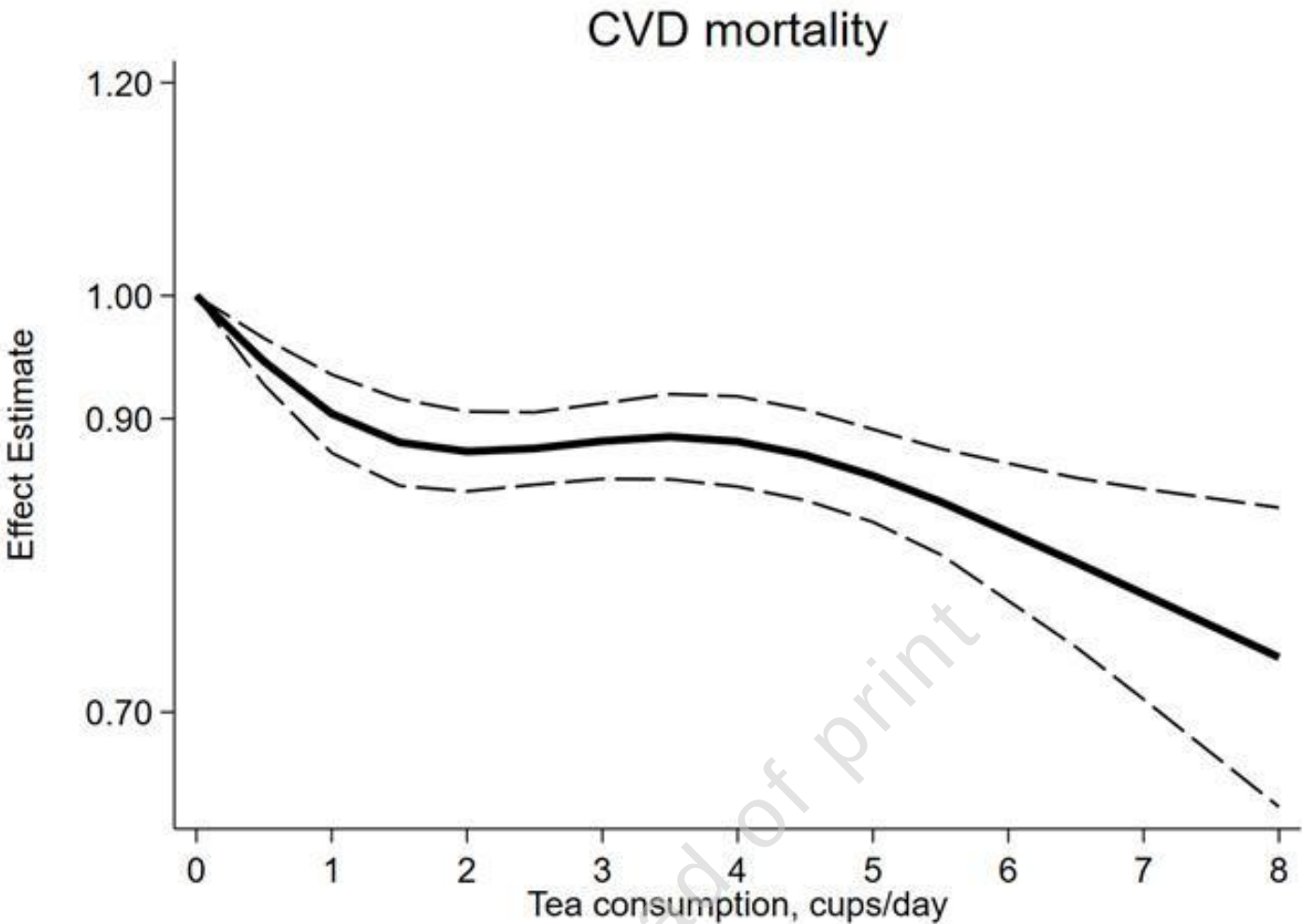


Figure 5



Cups/day	ES	95% CI
0	1.00 (reference)	
0.5	0.95	(0.93-0.96)
1	0.90	(0.87-0.93)
1.5	0.88	(0.85-0.92)
2	0.88	(0.85-0.91)
2.5	0.88	(0.85-0.90)
3	0.88	(0.85-0.91)
3.5	0.89	(0.85-0.92)
4	0.88	(0.85-0.92)
4.5	0.87	(0.84-0.91)
5	0.86	(0.82-0.89)
5.5	0.84	(0.80-0.88)
6.5	0.80	(0.74-0.86)
7	0.77	(0.71-0.85)
8	0.73	(0.65-0.83)
$P_{\text{nonlinearity}}$		0.0001

Figure 6

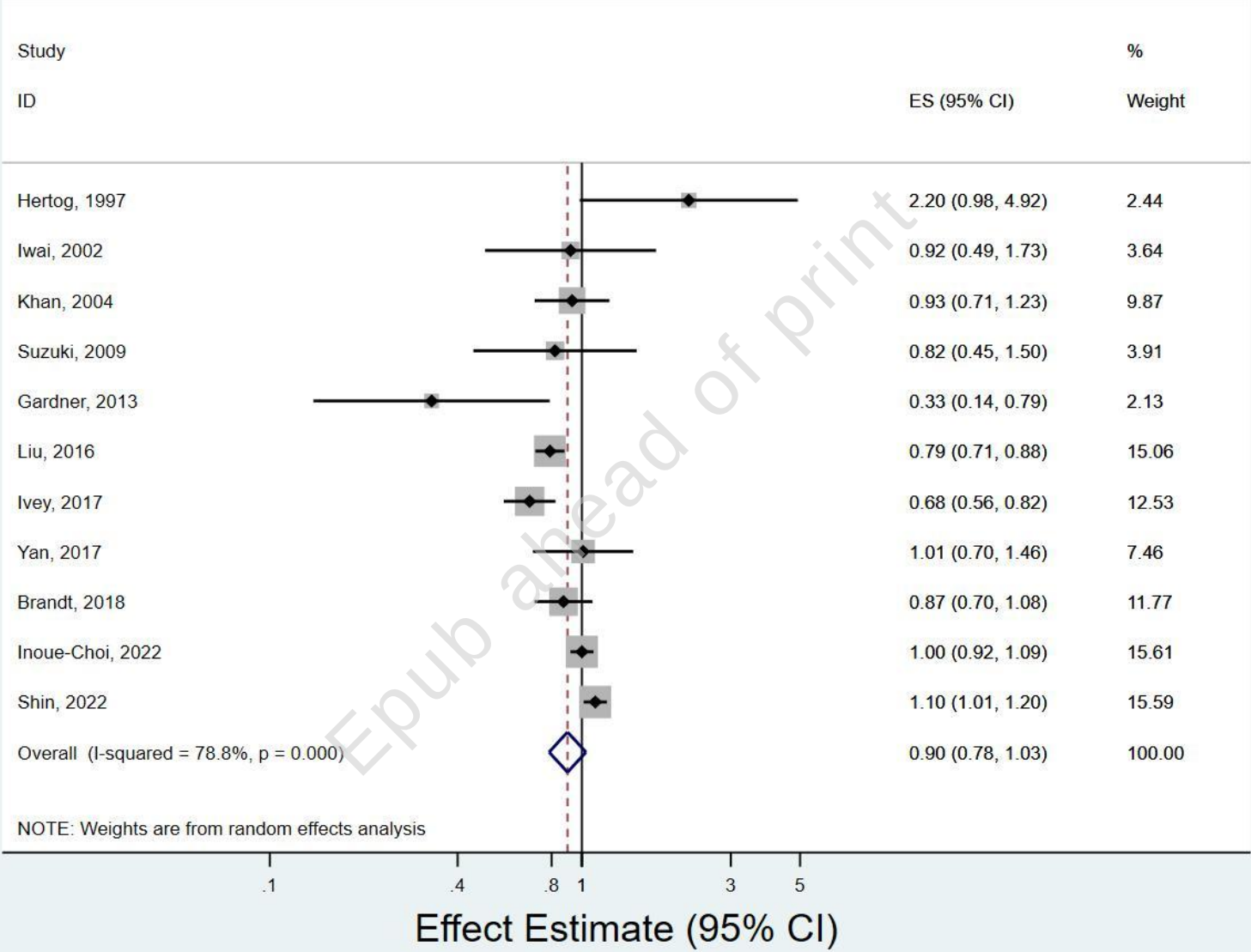
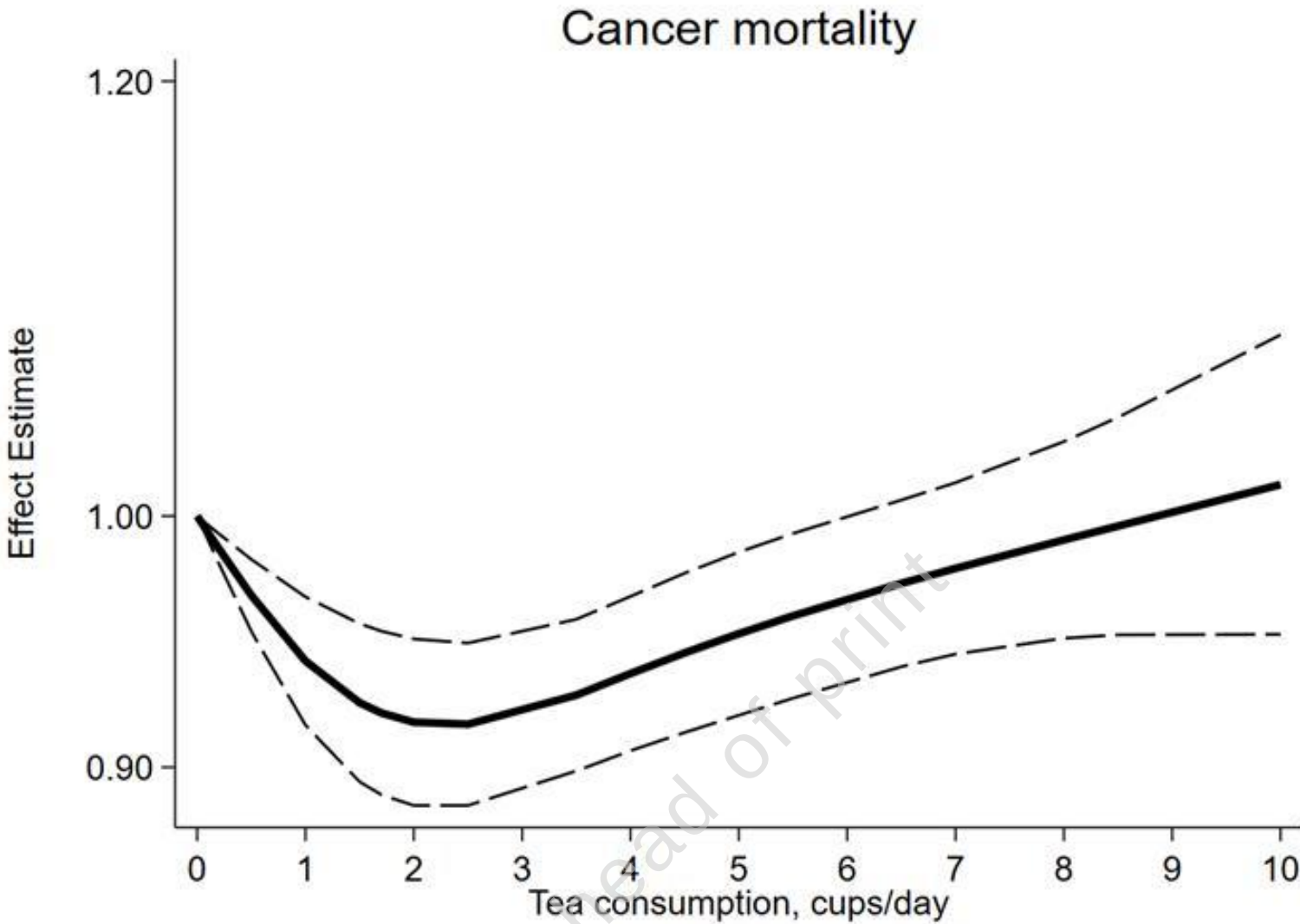


Figure 7



Cups/day	ES	95% CI
0	1.00 (reference)	
0.5	0.97	(0.95-0.98)
1	0.94	(0.92-0.97)
1.5	0.92	(0.89-0.96)
2	0.92	(0.89-0.95)
2.5	0.92	(0.89-0.95)
3.5	0.93	(0.90-0.96)
4	0.94	(0.91-0.97)
4.5	0.94	(0.91-0.98)
5	0.95	(0.92-0.98)
5.5	0.96	(0.93-0.99)
6.5	0.97	(0.94-1.01)
7	0.98	(0.94-1.01)
8	0.99	(0.95-1.03)
8.5	1.00	(0.95-1.04)
10	1.01	(0.95-1.08)
10.5	1.02	(0.95-1.09)
$P_{\text{nonlinearity}}$		<0.0001